Ather Nawaz

AI Data Engineer · Senior Data Engineer · dbt · Databricks · RAG · Python

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PROFILE

AI data engineer with seven years of building and running a company's entire data platform from zero. In production I run Databricks, dbt and Python pipelines with statistical anomaly detection and alerting that catch problems before anyone opens a report. In my own open-source work I build LLM systems on top of data: Pipeline Doctor diagnoses failed dbt runs with lineage, RAG and an LLM, and quarantine-dq keeps bad records out of pipelines without losing them. I care most about the data an LLM is given, because that is where most AI problems start.

SKILLS

Languages: Python, SQL, PySpark, Java
Transformation & orchestration: dbt, Azure Data Factory, Great Expectations; Airflow and Dagster in personal projects
Platforms & storage: Databricks, Delta Lake, Unity Catalog, Apache Iceberg, DuckDB, Snowflake, SQL Server, PostgreSQL
Cloud & DevOps: Azure (Data Factory, ADLS Gen2, Databricks), AWS (S3, Glue, Athena, Lambda, SQS, CloudWatch), Docker, Terraform (personal projects), GitHub Actions
AI / LLM data: RAG pipelines, embeddings, pgvector, Chroma, Pinecone, LlamaIndex, LangChain, LLM APIs (OpenAI, Anthropic)
BI & APIs: Power BI, FastAPI, REST, EDIFACT, SFTP integrations

EXPERIENCE

Senior Software Engineer (Data Engineering) · SCA Logistics & Fulfilment GmbH, Ingolstadt — Aug 2019 – present

- Joined in August 2019 as the first dedicated IT and data person; designed and built the company's entire data platform and still own it end to end, from ingestion to reporting.
- Ingest 300,000+ logistics records a day from the warehouse system, carriers and client shops; wrote the DHL bulk loader in Python (pyodbc, fast_executemany) into SQL Server.
- Run the lakehouse on Azure: Data Factory for orchestration, ADLS Gen2 storage, Databricks with Delta Lake, Auto Loader, PySpark and dbt in bronze/silver/gold layers, Unity Catalog for access control, and separate dev, test and prod environments deployed through GitHub Actions CI/CD.
- Added automated checks and alerting across the pipelines: no silent failures since 2022 and around 40% less time to spot data anomalies.
- Built analyses that changed decisions: order anomaly detection (+15% fulfilment accuracy), a shipping profitability model that exposed loss-making orders (10–12% better margins), and return-driver analysis (8% fewer returns).
- Rebuilt carrier invoice reconciliation: turned undocumented logic into 79 explicit DHL/DPD fee-code mappings and 12 pricing strategies, then replaced ~1,280 lines of manual SQL with a 13-step Python pipeline, a FastAPI backend and a dashboard, shipped in Docker.
- Built the integrations layer for fulfilment clients: an order API that normalises two e-commerce JSON formats (native and ChannelEngine) into one SQL Server schema, an EDIFACT ORDERS (D01B) parser into a three-table model, and a config-driven Python SFTP sync on Azure Blob Storage.
- Replaced hand-stitched exports from Shopify, Billbee and PlentyMarkets with one automated reporting layer and Power BI dashboards, cutting reporting time by about 80% for 12 business analysts.

Freelance Software Developer · Self-employed, Karachi / Germany — Jul 2017 – Jul 2019

- Java development for clients while relocating to Germany: e-commerce shop integrations and automated stock-level updates.

Java Software Engineer · Pacslink Corporation, Karachi — Jan 2015 – Jun 2017

- Built a hospital management and information system in Java (Spring, Hibernate, J2EE), from data model to deployment.
- Developed the IPD (in-patient), OPD (out-patient), pathology and blood bank modules, plus the finance and accounts modules.

Freelance IT Consultant · Self-employed, Karachi — Jun 2013 – Dec 2014

- Java projects for clients: a face-based attendance system, an HR system and image compression.

Senior IT Manager · Go Green Pakistan, Karachi Sep 2009 – May 2013

- Built the organisation's website and was responsible for all of its IT.

SELECTED PERSONAL PROJECTS

- **Pipeline Doctor:** LLM root-cause diagnosis for failed dbt runs: reads manifest and run results, traces lineage (NetworkX), retrieves the relevant SQL with RAG (LlamaIndex, pgvector) and explains the cause in plain English; Slack alerts via AWS Lambda.
- **Meridian:** Real-time e-commerce analytics: Kafka with Avro and Schema Registry, Flink (event-time watermarks, CEP fraud detection, exactly-once) into Iceberg, then dbt and DuckDB; monitored with Prometheus and Grafana; 30 tests.
- **quarantine-dq:** Python library that quarantines bad records instead of crashing or silently dropping them, and replays them after a fix; pandas, Spark, Airflow and dbt adapters, optional LLM failure classification; 82 passing tests.
- **dbt-contracts:** Live web app (dbt-contracts.vercel.app) that generates dbt data contracts with an LLM and enforces them in CI through a GitHub Action: FastAPI, React.

All code: github.com/ATHARNAWAZ

CERTIFICATIONS

Databricks Fundamentals (2026). In progress: Databricks Data Engineer Associate, dbt Analytics Engineering, Snowflake SnowPro Core.

EDUCATION

BSc Computer Science · Iqra University, Karachi 2010 – 2014

LANGUAGES

English (C2), German (B1, telc certified; used daily at work), Urdu (native)